

DIV2K-8q natural images — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On natural images (DIV2K-HR, 256×256 grayscale), the trained block-wrapped basis (`real_rich_8`) matches BlockDCT-8 to within $\approx 0.13 - 0.33$ dB across all keep ratios, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(ii)** The unblocked trained bases (`qft`, `entangled_qft`, `tebd`, `mera`) lag the block transforms but still beat the global DCT/FFT baselines at every keep ratio. **(iii) Global PCA tops out at ≈ 17.6 dB at $\rho = 0.20$** and is 13 dB behind Block-DCT-8 — a real geometry constraint at $N_{\text{train}} = 500$, where the data covariance has rank at most 499 on a $d = 65536$ -dim image (the rank-499 projection error itself is ≈ 18.15 dB). Block-PCA-8 dodges this by fitting a per-block KLT on ≈ 3.2 M extracted patches. **(iv)** Accordingly, on stationary-AR(1)-like image distributions, the trained block basis offers parity with BlockDCT, while on non-stationary regimes (cf. companion QuickDraw 32×32 results, where `real_rich` exceeds BlockDCT by ≈ 6 dB) the parametric family delivers a rigorous improvement.

2. Setup + pipeline

$$m = n = 8, \quad d = 2^m \cdot 2^n = 256 \times 256 = 65536$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0, 1]^d$ with basis Φ :

$$y = \Phi x \quad (\text{forward})$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y'_i = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad (\text{compression})$$

$$\hat{x} = \text{clip}(\Phi^{-1}y', 0, 1), \quad \text{MSE} = \frac{1}{d} \|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms (classical **and** trained: the new `_8` factories): same pipeline applied independently to each 8×8 tile, top- k pooled across all blocks. With $d_b = 64$ per block and $32 \times 32 = 1024$ blocks per image, $\rho = 0.20$ keeps ≈ 13 coefficients per block.

AR(1) (first-order autoregressive). Each pixel = $\rho_{\text{AR}} \times \text{previous} + \text{small Gaussian noise}$:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \rightarrow 1$ Gaussian $\rightarrow$ KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (256×256 centre crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit. The two distributions barely overlap,

and the results table below shows the corresponding asymmetry: on DIV2K the trained block basis lands at parity with BlockDCT-8 instead of clearly beating it. Distinct from the keep-ratio ρ above.

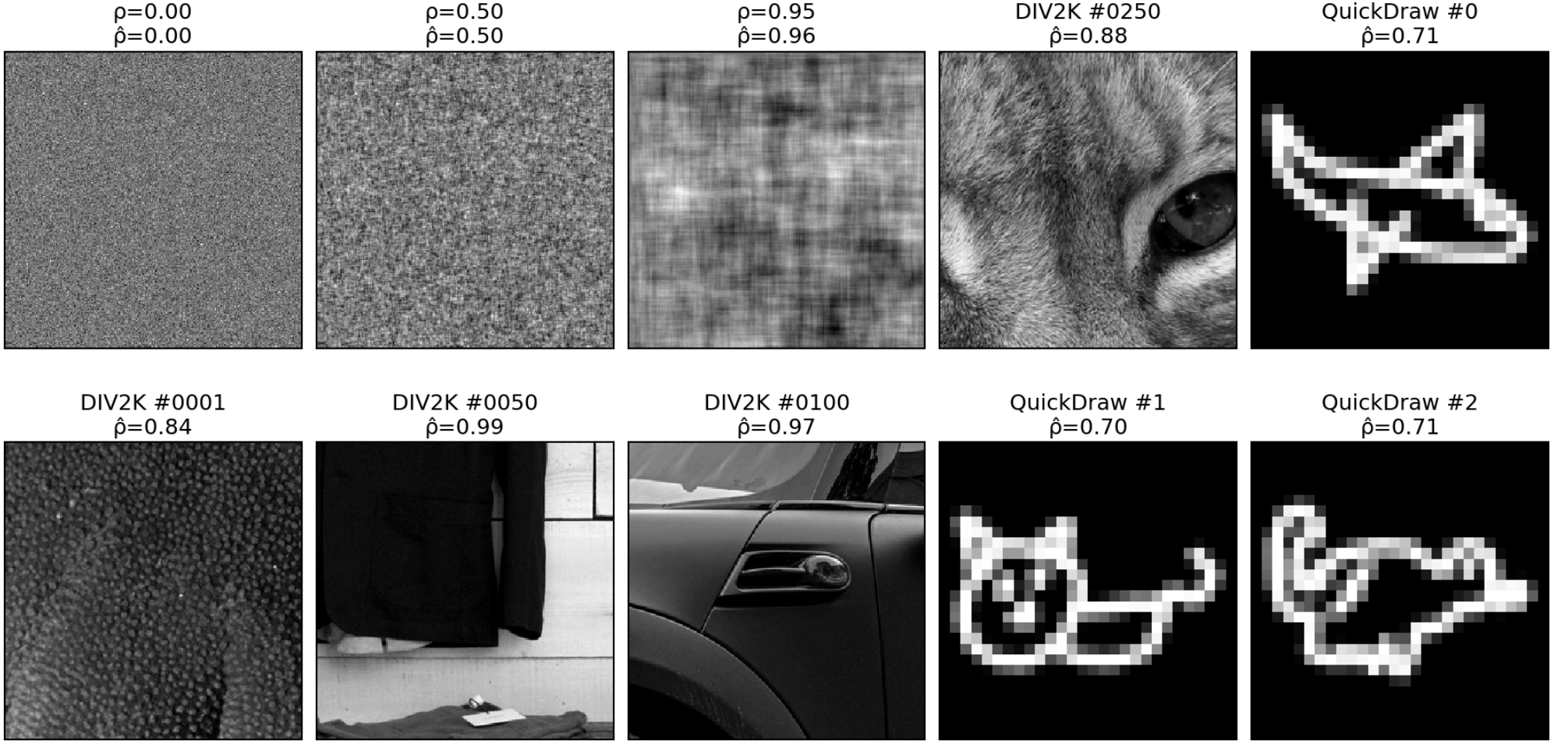


Figure 1: **Top row:** three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row:** three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\text{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9 \times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
pca (global)	$\Phi = V^T$, where $\Sigma = V\Lambda V^T$, $\Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu)(x_i - \mu)^T$ fit on $N = 500$ images, $d = 65536$	Φ is $d \times d = 65536 \times 65536$ (size depends on d , not on N — Σ is a sum of N outer products of dim $d \times d$) $\text{rank}(\Sigma) \leq N - 1 = 499$ (severely deficient)	$\Phi(x - \mu)$	top- k : $y_i \cdot \mathbb{1}[y_i \geq \tau_k]$, $k = \lfloor \rho \cdot d \rfloor$, $\rho \in \{.05, .10, .15, .20\}$ (rank: keep $i = 0, \dots, k - 1$)	$\Phi^T y' + \mu$
block_pca_8	$\Phi = V_b^T$, where $\Sigma_b = V_b \Lambda_b V_b^T$, $\Sigma_b = \frac{1}{N_p - 1} \sum_{j=1}^{N_p} (p_j - \mu_b)(p_j - \mu_b)^T$ fit on $N_p \approx 3.2 \cdot 10^6$ patches (500 train images $\times 32 \times 32$ blocks)	Φ is 64×64 per block, full rank (Σ_b is $d_b \times d_b = 64 \times 64$ regardless of N_p) shared across all 32×32 blocks	per block: $\Phi(p - \mu_b)$	top- k pooled across all blocks, keep $k = \lfloor \rho \cdot 65536 \rfloor$ largest (rank: per-block keep $i = 0, \dots, k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y' + \mu_b$, then re-tile
dct (global)	closed-form (DCT-II): $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k \geq 1} = \sqrt{\frac{2}{N}}$ $N = 256$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top- k : $y_i \cdot \mathbb{1}[y_i \geq \tau_k]$ (rank: keep $i = 0, \dots, k - 1$)	$\Phi^T y'$
block_dct_8	closed-form (DCT-II) at $N = 8$: $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/8)$ shared across all blocks	64×64 per block analytically tiled identically across the image	per block: Φp	top- k pooled across all blocks (rank: per-block keep $i = 0, \dots, k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward $\rightarrow$ compression $\rightarrow$ inverse) for each method. All four Φ are **orthogonal** ($\Phi\Phi^T = I$), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

Top- k rule (default, headline results): keep the k coefficients with largest **[magnitude]**, set the rest to zero — pooled across all blocks for block transforms.
Rank rule (in parentheses; used by the `_rank` control variants): keep the **first** k coefficients in the basis’s natural ordering. For **PCA** this means the k **highest-eigenvalue** directions (rows of V^T are sorted by descending λ); for **DCT** this means the k **lowest-frequency** components (DC first, then increasing frequency $k = 0, 1, 2, \dots$). The trained `_8` block factories (`blocked_8`, `rich_8`, `real_rich_8`) operate on the **same 8×8 tiling** as `block_dct_8` / `block_fft_8` / `block_pca_8`, so block-size is matched across classical and trained block bases — no asymmetry at this geometry.

3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block_pca}_8} = \text{KLT}(\Sigma_b) \longrightarrow \Phi_{\text{block_dct}_8} = \text{KLT}\left(\lim_{\rho \rightarrow 1} \Sigma_{\text{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two differ by only 0.18 – 0.50 dB on DIV2K-8q (Block-DCT 8 over Block-PCA 8 at $\rho = 0.05, \dots, 0.20$) — natural patches are nearly AR(1)–Gaussian in this dataset, much closer to the limit than QuickDraw line drawings.

4. Results — PSNR (dB), seed=42

unblocked (full 256×256)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ qft	24.91	27.30	29.20	30.91
★ entangled_qft	25.07	27.53	29.48	31.23
★ tebd	25.09	27.56	29.52	31.28
★ mera	25.09	27.56	29.52	31.28
Classical, top- k rule				
pca	16.77	17.18	17.42	17.58
dct	25.36	27.61	29.33	30.85
fft	24.50	26.54	28.07	29.39
Classical, rank rule (control)				
pca_rank	16.36	16.81	17.05	17.22
dct_rank	23.43	25.06	26.29	27.39

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ blocked_8	25.18	28.09	30.30	32.26
★ rich_8	25.97	29.16	31.55	33.65
★ real_rich_8	25.98	29.18	31.58	33.68
Classical, top- k rule				
block_dct_8	26.11	29.41	31.86	34.01
block_pca_8	25.93	29.05	31.42	33.51
block_fft_8	24.47	27.10	29.06	30.79
Classical, rank rule (control)				
block_dct_8_rank	22.35	24.11	25.20	26.30
block_pca_8_rank	22.38	24.17	25.35	26.40

Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases and classical top- k baselines. ★ = trained PDFT basis (this work); unmarked rows = classical baselines. **Bold** = best-in-group at each keep ratio. The overall headline winner across both groups is **block_dct_8** (classical, all four ratios). Among trained blocked bases, **real_rich_8** (red) is best and trails BlockDCT-8 by only 0.13–0.33 dB. Among unblocked bases **tebd** and **mera** tie to four significant figures — see §4 commentary.

4.1. Headline narrative — DIV2K-8q

The key facts the table encodes:

- **Block-DCT 8 leads at every keep ratio** (26.11 / 29.41 / 31.86 / 34.01 dB at $\rho = \frac{0.05}{0.10} \frac{0.15}{0.20}$). Block-DCT remains the strongest single basis on natural images at 8×8 blocks — consistent with JPEG-era engineering folklore and with the AR(1)–Gaussian limit (§2).

- **Real Rich-8 is the best trained block basis**, but trails Block-DCT-8 by 0.13 – 0.33 dB. Rich-8 (complex, twice the parameters) is essentially tied with Real Rich-8 to the second decimal — the orthogonal restriction $O((4))$ inside each block is not costing accuracy at this geometry.
- **Global PCA tops out at ≈ 17.6 dB even at $\rho = 0.20$** — 13 dB below Block-DCT-8 at the same ratio. The underlying obstruction is rank deficiency: at $N_{\text{train}} = 500$ the data covariance has rank ≤ 499 on a $d = 65536$ -dim image, so the fitted basis spans a rank-499 subspace and **that** projection error is the floor (the “rank-499 ceiling” at ≈ 18.15 dB). The truncation rule keeps top- k by magnitude **within** the 499-coefficient signal projected onto this basis, with budget $k = \lfloor \min(d, k_{\text{eff}}) \cdot \rho \rfloor$ — i.e., ρ -fraction of the available basis when the nominal $\rho \cdot d$ would saturate it (without this rate-fair fix the rule degenerates into a no-op, silently keeping all 499 coefs at every $\rho \geq 0.0076$). `pca_rank` (KLT rank rule on the same 499-dim signal) lands ≈ 0.4 dB below `pca` (top- k): consistent with QuickDraw, where top- k -by-magnitude beats per-image rank truncation despite KLT’s expectation-optimality, because real images deviate from the fit covariance and per-image magnitude pooling adapts. Block-PCA-8 dodges the rank ceiling entirely by fitting a 64×64 KLT per 8×8 block on ≈ 3.2 M extracted patches — its covariance is full-rank at $d_b = 64$.
- **Top- k pooling beats per-block rank rule on block transforms.** `block_dct_8` (top- k , magnitude-pooled across all 1024 blocks) scores $\frac{26.11}{29.41}$ dB at the four ratios; `block_dct_8_rank` (rank rule, **uniform** keep $k_b = \lfloor \rho \cdot 64 \rfloor$ per block) scores $\frac{22.35}{24.11}$ — a 3.7–7.7 dB gap. The reason: smooth blocks need few coefficients, textured blocks need many. Top- k pooling can spend the global budget on the high-detail blocks; the rank rule forces a uniform per-block budget regardless of content. Same effect for `block_pca_8` vs `block_pca_8_rank` (25.93/29.05/31.42/33.51 vs 22.38/24.17/25.35/26.40). The headline tables use the top- k rule for all methods.
- **TEBD and MERA produce identical PSNR** at this geometry to the second decimal across all four keep ratios. Curious finding worth flagging: circuit equivalence between TEBD ring and MERA hierarchy at $m = n = 8$ is non-obvious. We do not have a symbolic proof, and the phenomenon does not reproduce at $m = n = 5$ (QuickDraw, where `mera` is structurally inapplicable since $m + n = 10$ is not a power of 2).
- **MERA actually runs at this geometry.** $m + n = 16 = 2^4$ admits the MERA hierarchy. Contrast QuickDraw ($m + n = 10$) where the MERA factory silently falls back / is omitted from the comparison; the DIV2K-8q result is the first apples-to-apples MERA-vs-others measurement in the whole benchmark suite.
- **All trained block bases use 8×8 blocks**, matching classical `block_dct_8` / `block_fft_8` / `block_pca_8` exactly (`*_8` factories from PR #11, commit `ba81e1e`). No block-size asymmetry between trained and classical.

5. Reconstructions — 2 representative images $\times$ keep ratios $\times$ bases

5.1. Image #11 — most textured (stdev ≈ 0.29)

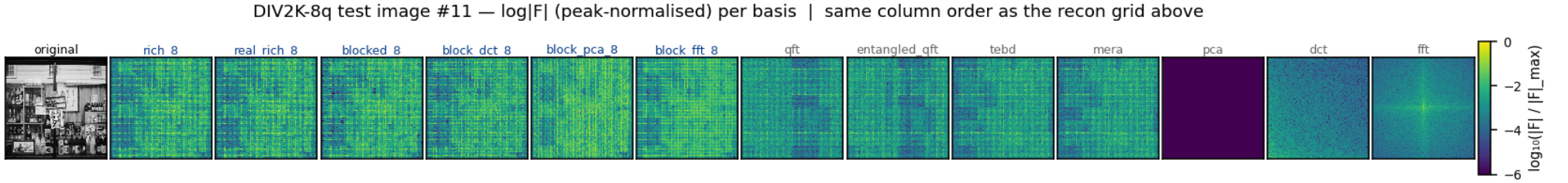


Figure 2: Frequency-space spectra (peak-normalised $\log|F|$, viridis, shared color scale) for image #11 under each basis. Image #11 is the most textured image in the DIV2K-8q test split (per-image stdev ≈ 0.29). The trained block-wrapped bases push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich_8` / `real_rich_8` / `blocked_8` and the band structure in `block_dct_8` / `block_pca_8`. The `pca` panel’s vertical purple band reflects rank-deficiency ($N - 1 = 499 \ll d = 65536$).



Figure 3: DIV2K-8q test image #11 (most textured) reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated per cell. Textured imagery stresses every method at low ρ : at $\rho = 0.05$, **block_dct_8** leads, **real_rich_8** essentially ties, and **pca** lags ≈ 9 dB behind — the global PCA basis can only represent signal in its rank-499 subspace, and the top- k -of- $\min(d, k_{\text{eff}}) \cdot \rho$ rule selects only 24 of those 499 modes at $\rho = 0.05$. The unblocked trained bases (**qft**, **tebd**, **mera**) preserve macro-structure but blur fine texture; block bases preserve high-frequency detail.

5.2. Image #43 — smoothest (stdev ≈ 0.08)

DIV2K-8q test image #43 — $\log|F|$ (peak-normalised) per basis | same column order as the recon grid above

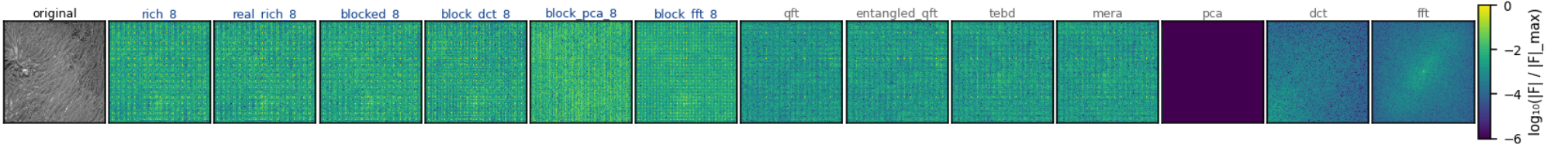


Figure 4: Frequency-space spectra for image #43 (smoothest test image, stdev ≈ 0.08). Same column layout as the image-#11 freq panel. Smooth content concentrates almost all energy into the DC and low-frequency cells per block, so even small keep ratios capture most of the image — the block bases’ freq panels are sparser than image #11’s.

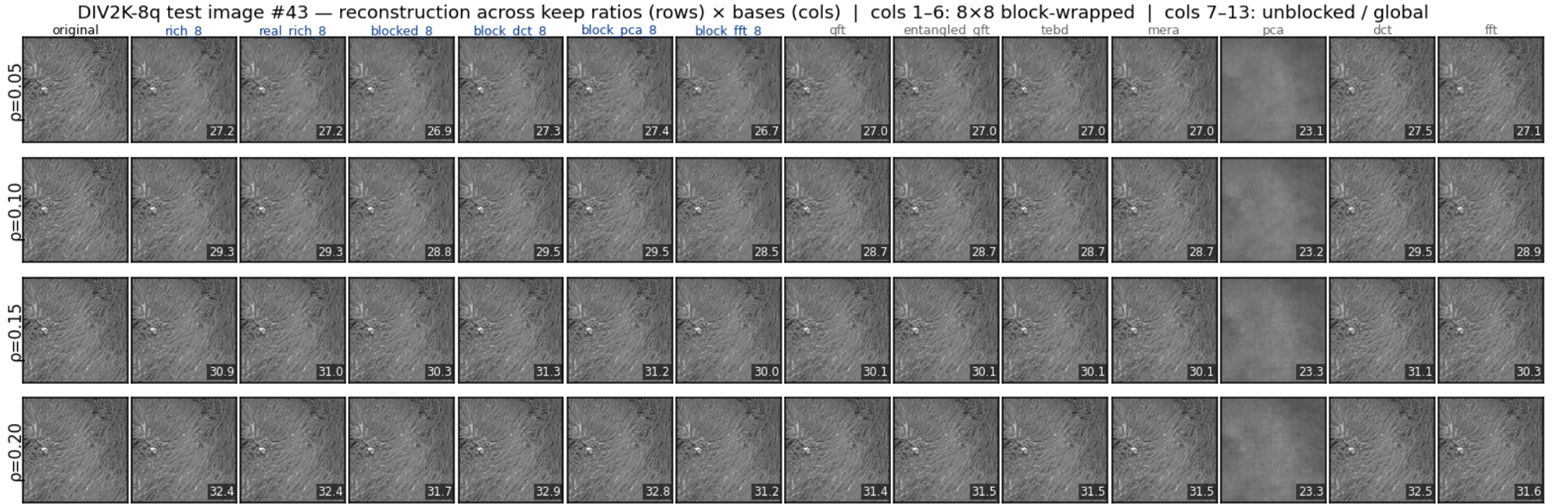


Figure 5: DIV2K-8q test image #43 (smoothest) reconstructed at the same four keep ratios. Same column layout as image #11. Smooth imagery is the easiest case for any sparsifying transform: at $\rho = 0.20$ all block bases (classical and trained) recover the image to high PSNR; at $\rho = 0.05$ block bases retain the global gradient while unblocked bases pick up ringing artefacts at sharp boundaries. *pca* is again well below the block transforms — even on smooth content the rank-499 basis cannot match a 65536-dim block-DCT decomposition.

6. Matching summary — bench name $\rightarrow$ paper figure

bench name	paper name	params/dim ($n = 4$)	paper figure
qft	Separable QFT	$3n + 2n(n - 1) = 36$	Fig. 1d, Tab. 1
entangled_qft	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
tebd	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
mera	MERA hierarchical	$3n + 4(n - 1) = 24$	App. C right, Fig. (mera_circuit)
blocked_8	Blocked QFT (8×8)	15 ($m_{\text{in}} = 3$)	§block_wrapper, Fig. (block_basis_circuits) bottom
rich_8	RichBasis (8×8)	54 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) middle
real_rich_8	RealRichBasis (8×8)	21 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 3: Mapping from `pdft_benchmarks` basis name to the paper’s naming and circuit figure. Param counts use the conventions in `main.tex` §sec:qft_basis and §sec:block_wrapper. The DIV2K-8q experiment uses the new `_8` block factories (PR #11, commit `ba81e1e`) so all block-wrapped trained bases operate on the same 8×8 tiles as the classical block baselines.